

Autodesk® Scaleform®

ActionScript 2 XML

Scaleform 4.2 ActionScript 2 XML

: Prasad Silva
: 3.0
: July 28, 2010

Copyright Notice

Autodesk® Scaleform® 4.2

© 2012 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AUGI, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo) Built with ObjectARX (design/logo), Burn, Buzzsaw, CAiCE, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, DesignServer, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWFx, DXF, Ecotect, Evolver, Exposure, Extending the Design Team, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, Freewheel, GDX Driver, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, RiverCAD, Robot, Scaleform, Scaleform GfX, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, Tinkerbox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

XML

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

(301) 446-3199

.....	1
1. XML	2
1.1 ActionScript XML	2
1.2 C++	2
1.3 C++ DOM	3
1.4	4
2.	6
3.	8
3.1 ActionScript XML	8
3.2 ActionScript XML.status	9
3.3 ActionScript XML.docTypeDecl	9

Autodesk® Scaleform® SDK

Flash

PC

Scaleform

Adobe® Flash® Studio

Flash

AS2 XML API

ActionScript 2 XML

XML

XMLNode

DOM

XML

Scaleform 4.1

Scaleform

DOM

SAX2

XML

Scaleform Player

Expat

Scaleform

XML

Windows®, Linux®, MacOS®, Xbox 360®, PSP® (PlayStation®Portable), PlayStation®2 (PS2), PlayStation®3 (PS3™), Nintendo® Wii

Scaleform XML

1. XML

Scaleform XML

1.1 ActionScript XML

ActionScript XML

Scaleform

XML

FxPlayer.cpp

```
...
#include "XML/XML_Expatri.h"
using namespace Scaleform;
...
Ptr<XML::Parser> pexpatXmlParser = *new XML::ParserExpat;
Ptr<XML::SupportBase> pxmlSupport = *new XML::Support(pexpatXmlParser);
mLoader.SetXMLSupport(pxmlSupport);...
```

1.2 C++

C++

XML

ActionScript XML

XML

DOM

Scaleform

```
...
#include "GFx/XML/XML_DOM.h"
using namespace Scaleform;
...
//          DOM
XML::DOMBuilder domBuilder(Loader.GetXMLSupport());
//          DOM
XML::DOMBuilder domBuilder2(Loader.GetXMLSupport(), true);
...
//  xml      DOM
Ptr<XML::Document> pdoc = domBuilder.ParseFile("inputfile.xml",
                                              mLoader.GetFileOpener());

//  xml      DOM
Ptr<XML::ObjectManager> pObjMgr = *new XML::ObjectManager();
Ptr<XML::Document> pdoc2 = domBuilder.ParseFile("inputfile.xml",
                                              mLoader.GetFileOpener(), pObjMgr);
...
```

ActionScript XML

```

...
#include "GFx/XML/XML_DOM.h"
#include "GFx/XML/XML_Expatri.h"
using namespace Scaleform;
...
Ptr<XML::Parser> pexpatXmlParser = *new XML::ParserExpat;
Ptr<XML::Support> pxmlSupport = *new XML::Support(pexpatXmlParser);
...

//          DOM
XML::DOMBuilder domBuilder(pxmlSupport);
//          DOM
XML::DOMBuilder domBuilder2(pxmlSupport, true);
...
//
...

```

1.3 C++ DOM

DOM	Scaleform
-----	-----------

```

...
using namespace Scaleform;
...
void PrintDOMTree(XML::Node* proot)
{
    switch (proot->Type)
    {
        case XML::ElementNodeType:
        {
            XML::ElementNode* pnode =
                static_cast< XML::ElementNode*>(proot);
            if (pnode->Prefix.GetSize() > 0)
            {
                printf("ELEM - '%s:%s' ns:'%s' prefix:'%s'"
                    " localname: '%s'",
                    pnode->Prefix.ToCStr(),
                    pnode->Value.ToCStr(),
                    pnode->Namespace.ToCStr(),
                    pnode->Prefix.ToCStr(),
                    pnode->Value.ToCStr());
            }
        }
        else
        {

```

```

        printf("ELEM - '%s' ns:'%s' prefix:''"
               " localname: '%s'",
               pnode->Value.ToCStr(),
               pnode->Namespace.ToCStr(),
               pnode->Value.ToCStr());
    }
    for (XML::Attribute* attr = pnode->FirstAttribute;
         attr != NULL; attr = attr->Next)
    {
        printf(" {%s,%s}", attr->Name.ToCStr(),
               attr->Value.ToCStr());
    }
    printf("\n");
    for (XML::Node* child = pnode->FirstChild; child != NULL;
         child = child->NextSibling)
        PrintDOMTree(child);
    break;
}
case TextNodeType:
{
    printf("TEXT - '%s'\n", pnode->Value.ToCStr());
    break;
}
default:
{
    printf("UNKN\n");
}
}
}
...
Ptr<XML::Document> pdoc = domBuilder.ParseFile("inputfile.xml",
                                              Loader.GetFileOpener());
PrintDOMTree(root);

```

1.4

```

Scaleform                                XML_DOM.cpp                                XML
XML      DOM
//
//
#ifdef SF_BUILD_DEBUG
////
//      DOM

```



```
// #define SF_XML_DOCBUILDER_TRACE
//
//      DOM
// (
//
// #define SF_XML_DOCBUILDER_DOM_TREE_DUMP
//
#endif // #ifdef SF_BUILD_DEBUG
```

2.

Scaleform		XML
GFx::XML::Parser	Scaleform XML	XML_Support.h
XML_Support.h	GFx::XML::ParserHandler	SAX2

```
namespace Scaleform { namespace GFx { namespace XML {
// XML
//
//      /
//
// XML
class Parser : public RefCountBaseNTS<Parser, Stat_Default_Mem>
{
public:
    virtual ~Parser() {}

    // Parse methods

    virtual bool    ParseFile(const char* pfilename, FileOpenerBase* pfo,
                             ParserHandler* pphandler) = 0;
    virtual bool    ParseString(const char* pdata, UPInt len,
                               ParserHandler* pphandler) = 0;
};
}}} //SF::GFx::XML
```

XML	DOM	GFx::XML::ParserLocator
ParseFile	ParseString	
GFx::XML::ParserHandler		

```
namespace Scaleform { namespace GFx { namespace XML {
// SAX2
// DOM          SAX2
//
//
//
class ParserHandler : public RefCountBase<ParserHandler, StatMV_XML_Mem>
{
public:
    ParserHandler() {}
    virtual ~ParserHandler() {}
}
```

```

//
virtual void      StartDocument() = 0;
virtual void      EndDocument() = 0;

//
virtual void      StartElement(const StringRef& prefix,
                               const StringRef& localname,
                               const ParserAttributes& atts) = 0;
virtual void      EndElement(const StringRef& prefix,
                             const StringRef& localname) = 0;

//
virtual void      PrefixMapping(const StringRef& prefix,
                                const StringRef& uri) = 0;

//
virtual void      Characters(const StringRef& text) = 0;

//
virtual void      IgnorableWhitespace(const XMLStringRef& ws) = 0;

//
virtual void      SkippedEntity(const XMLStringRef& name) = 0;

// GfX::XML::Parser                                GfX::XML::ParserHandler
// // //                                           xml
virtual void      SetDocumentLocator(const ParserLocator* plocator) = 0;

//
virtual void      Comment(const StringRef& text) = 0;

//
virtual void      Error(const ParserException& exception) = 0;
virtual void      FatalError(const ParserException& exception) = 0;
virtual void      Warning(const ParserException& exception) = 0;

};
}}} //SF::GfX::XML

```

3.

3.1 ActionScript

XML

ActionScript	XMLNode	XML		
	ActionScript		DOM	
ActionScript	DOM		ActionScript	
XML.load		XMLNode		DOM
ActionScript	XMLNode	DOM		
		XML	XML	XMLNode

```
//      DOM
XMLNode temp = root.firstChild;
//
temp.someProperty = someValue;
//
XMLNode temp = temp.nextSibling;
//
temp = temp.prevSibling;
//
trace(temp.someProperty)
```

Flash™		Scaleform		Scaleform
XMLNode				

	XMLNode		ActionScript
XMLNode			

```
//      DOM
XMLNode temp = root.firstChild;
//
temp.attributes.someProperty = someValue;
//
XMLNode temp = temp.nextSibling;
//
temp = temp.prevSibling;
//
trace(temp.attributes.someProperty)
```

Scaleform

3.2 *ActionScript XML.status*

ActionScript 2.0 XML

ActionScript

- 0
- -2 A CDATA
- -3 XML
- -4 DOCTYPE
- -5
- -6 XML
- -7
- -8
- -9
- -10

ActionScript

```
//  
// XML_ERROR_NONE = 0  
// XML_ERROR_INVALID_TOKEN = 0 (ie: XML.parse("http://www.google.com"))  
// XML_ERROR_UNCLOSED_CDATA_SECTION = -2  
// XML_ERROR_UNCLOSED_TOKEN = -3  
// XML_ERROR_INVALID_TOKEN = -4  
// XML_ERROR_INVALID_TOKEN = -5  
// XML_ERROR_UNCLOSED_TOKEN = -8  
// XML_ERROR_NO_ELEMENTS = -9  
// XML_ERROR_TAG_MISMATCH = -10  
//
```

GfX::XML::DOMBuilder

XML

GfX::XML::ParserLocator

3.3 *ActionScript XML.docTypeDecl*

Flash®	“ ActionScript XML		DOCTYPE
XML.docTypeDecl	DTD	”	Flash Scaleform